

ENUMERATION LAB REPORT

BACKGROUND	
Course:	CIS 170 - Hands-On Ethical Hacking and Network Defense
Term:	Spring 2026
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Submission Platform:	Canvas
Report Date:	2026-01-29
Report Version:	1.0
Lab Title:	Enumeration
Module Reference:	Enumeration
Lab Type:	Live Virtual Machine Lab
Date Completed:	2026-01-29
Time Spent:	Approximately 1 hour
ENVIRONMENT	
Platforms and Systems:	
- Primary environment: Practice Labs / Live Virtual Machine Labs controlled sandbox. - Primary systems available in the topology: PLABDC01 (Windows Server 2019 - Domain Controller), PLABDM01 (Windows Server 2019 - Domain Member), PLABWIN10 (Windows 10 - Domain Member), and PLABKALI01 (Kali 2019.2 - Linux Kali Workstation). - The work centered mainly on PLABWIN10 for Windows-based enumeration tools (SuperScan, Hyena, Softterra LDAP Administrator, SolarWinds IP Network Browser / MIB Viewer) and PLABKALI01 for Linux-based tools (host, dnsrecon, dnsenum, rpcclient, nmap, nbtscan, wafw00f), with enumeration targeting the full lab topology and domain. - Remote access to the virtual machines was provided through the course platform.	
Context and Constraints:	
- The module focused on enumeration concepts, NetBIOS/SNMP/LDAP/DNS/SMB/Windows/Linux enumeration techniques, and prevention methods using both Windows and Kali Linux tools. - Practice Labs flagged the module as legacy content using deprecated software, so behavior could vary and some steps were best interpreted as guided sandbox demonstrations. - Windows Updates were disabled in the Practice Labs environment, so software behavior reflected a controlled legacy setup rather than a current hardened enterprise baseline. - The module aligned most closely with exam objective 5.2 (Information Security Assessment Methodologies).	
Authorization and Scope:	
- All documented activity took place only inside approved academic systems and isolated practice-lab sandboxes. - No public IP space, real organizations, or unauthorized targets were involved. - No production infrastructure, third-party systems, or live services were contacted. - The actions documented here were performed for defensive understanding, reconnaissance analysis, and ethical hacking education within the course environment. - Restricted material, live credentials, and personal data are omitted or redacted from this report.	
Work Objectives:	
- Review the basic components of enumeration, including NetBIOS, SNMP, LDAP, DNS, SMB, Windows, and Linux host enumeration. - Use Windows-based tools (SuperScan, Hyena, Softterra LDAP Administrator, IP Network Browser) on PLABWIN10 to extract usernames, groups, shares, services, and directory information. - Use Linux-based tools (host, dnsrecon, dnsenum, rpcclient, nmap, nbtscan, wafw00f) on PLABKALI01 to perform DNS, Windows host, website, and SMB enumeration.	
Enumeration Concepts and Defensive Framing:	
- The lab identified three core phases of enumeration: active connection establishment, information extraction, and targeted reconnaissance of services and accounts. - Enumeration types reviewed included NetBIOS, SNMP, LDAP, DNS, SMB, Windows host, and Linux host enumeration, with tools demonstrating both GUI and CLI approaches. - Tools demonstrated both Windows GUI utilities and Kali Linux CLI tools, showing how enumeration reveals usernames, groups, shares, services, DNS records, and directory data. - Additional methods reviewed included prevention techniques (WAF detection, SNMP service hardening, account lockout policies, DNS zone transfer restrictions, and service disabling), reinforcing that enumeration targets people, trust, and misconfigurations rather than only technology exploits.	

TECHNICAL ANALYSIS

Windows-Based Enumeration Tools on PLABWIN10:

- On PLABWIN10, SuperScan was launched, the Windows Enumeration tab selected, target 192.168.0.1 entered, and Enumerate clicked to extract NetBIOS information, users, groups, shares, and services; a full network scan (192.168.0.1-192.168.0.254) was also performed with the Scan tab.
- Hyena was installed and launched, PRACTICELABS.COM domain added, Domain Controllers expanded, and Services, Sessions, and Disk Space nodes double-clicked to view running services, active sessions, and disk usage.
- Softerra LDAP Administrator was installed, a new PLAB profile created for PLABDC01, authenticated as the current user, DC=PRACTICELABS/CN=Users expanded, and Quick Search used for "administrator"; the Administrator account password was reset via All Tasks.
- SolarWinds Toolset (IP Network Browser / MIB Viewer) was launched, SNMP community set to "public" on PLABDC01, and MIB Table downloaded to enumerate system information.

Linux-Based Enumeration Tools on PLABKALI01:

- On PLABKALI01, DNS enumeration used host -t ns / -t mx practicelabs.com, a custom loop with plab.txt services, host -l for zone transfer, dnsrecon -d practicelabs.com -t axfr, and dnsenum practicelabs.com.
- Windows host enumeration used rpcclient 192.168.0.3 -U admin (with Passw0rd), then srvinfo, lookupnames admin, enumprivs, and lsaenumsid.
- Linux host enumeration used nmap -O intranet, nmap -sV intranet, find / -perm -4000 -type f 2>/dev/null, and ps aux | grep root.
- Website enumeration used nmap -script=http-enum intranet; SMB enumeration used nmap -v -p 139,445 intranet, nbtscan -r 192.168.0.0/24, and nmap -script=smb-security-mode.
- Prevention testing used wafw00f http://intranet to detect a Web Application Firewall.

KEY COMMANDS, TOOLS, AND ARTIFACTS BY SYSTEM

PLABWIN10 — Windows Enumeration Tools:

SuperScan (Windows Enumeration tab)
 192.168.0.1 (Enumerate)
 Scan tab: 192.168.0.1-192.168.0.254
 Hyena (Domain: PRACTICELABS.COM)
 Services / Sessions / Disk Space
 Softterra LDAP Administrator (PLAB profile)
 Quick Search: administrator
 Reset Password: Passw0rd
 MIB Viewer: 192.168.0.1 / public

PLABKALI01 — Linux Enumeration Tools:

```
host -t ns practicelabs.com
host -t mx practicelabs.com
host -l practicelabs.com plabdc01.practicelabs.com
dnsrecon -d practicelabs.com -t axfr
dnsenum practicelabs.com
rpcclient 192.168.0.3 -U admin
srvinfo
lookupnames admin
enumprivs
lsaenumsid
nmap -O intranet
nmap -sV intranet
nmap -script=http-enum intranet
nmap -v -p 139,445 intranet
nbtscan -r 192.168.0.0/24
nmap -script=smb-security-mode intranet
wafw00f http://intranet
```

Primary Tools and Artifacts Observed:

- SuperScan NetBIOS/users/shares output, Hyena services/sessions/disk views, Softterra LDAP Administrator user search and password reset, MIB Viewer system table, host/dnsrecon/dnsenum DNS results, rpcclient srvinfo/SID/privs output, Nmap OS/service/website/SMB scans, nbtscan NetBIOS table, and wafw00f WAF detection.
- The most important evidence of enumeration activity in the lab was the detailed user/group/share data from SuperScan/Hyena, LDAP user details and password reset, DNS records and zone transfer attempts, rpcclient domain information, and Nmap script outputs revealing open services and security configurations.

DEFENSIVE MAPPING AND SUMMARY

Technique Summary:

- The completed work covered enumeration concepts, Windows tools (SuperScan, Hyena, LDAP Administrator, IP Network Browser), Linux tools (DNS enumeration, rpcclient, Nmap scripts, nbtscan), and prevention techniques (WAF detection, SNMP hardening, account lockout, DNS zone restrictions, and service disabling).
- The recurring theme across the module was that enumeration provides the attacker with detailed account, service, and configuration data before any exploitation occurs.

MITRE ATT&CK Mapping:

- Reconnaissance: T1590 Gather Victim Network Information - DNS, NetBIOS, and SMB enumeration directly mapped to network information gathering.
- Discovery: T1087 Account Discovery - rpcclient, Hyena, and Softterra LDAP Administrator demonstrated discovery of users, groups, and domain accounts.
- Discovery: T1049 System Network Connections Discovery / T1135 Network Share Discovery - SuperScan and nbtscan revealed shares and network services.
- Discovery: T1592 Gather Victim Host Information - Nmap OS detection and service versioning provided detailed host and software information.

Detection Opportunities:

- Unusual LDAP queries, SNMP GET requests, DNS zone transfer attempts, and repeated RPC calls are strong indicators of enumeration activity.
- IDS/IPS signatures for Nmap scripts, rpcclient connections, or wafw00f probes can trigger alerts.
- Log analysis of DNS queries, failed zone transfers, account lockout events, and firewall denies for ports 139/445/161 can identify reconnaissance early.

Defensive Controls:

- Disable unnecessary services (e.g., SNMP, Print Spooler), restrict zone transfers to authorized servers only, and enforce account lockout policies.
- Deploy Web Application Firewalls, use SNMPv3 with strong community strings, and enable logging for LDAP and RPC traffic.
- Implement network segmentation, host-based firewalls, and continuous monitoring to detect and block enumeration attempts.
- Apply least-privilege principles and regular security assessments to reduce the information exposed during the reconnaissance phase.

Completion Verification:

- Work was completed across conceptual enumeration review, Windows GUI tools, Linux CLI tools, and multiple prevention configurations.
- Completion was evidenced by successful NetBIOS/LDAP/SNMP/DNS/SMB enumeration results, rpcclient domain queries, Nmap script outputs, and WAF detection.
- The actual commands, tool configurations, and search targets used in the lab are preserved directly in this report for accurate documentation and reproducibility.

Key Findings:

- 1) Enumeration is the critical step that turns passive reconnaissance into actionable intelligence by revealing usernames, groups, shares, services, and DNS records.
- 2) Both Windows GUI tools (SuperScan, Hyena, LDAP Administrator) and Kali CLI tools (rpcclient, Nmap scripts) provide complementary visibility into the same environment.
- 3) SMB, LDAP, and SNMP remain high-value targets because default or weak configurations expose significant information with minimal effort.
- 4) Prevention requires layered controls - service hardening, zone transfer restrictions, account lockout policies, and WAF deployment - not just a single fix.
- 5) The strongest defensive takeaway was that early detection and restriction of enumeration activity is one of the most effective ways to stop attacks before exploitation begins.

Issues / Deviations / Notes:

- The lab used deprecated software and a legacy environment, so exact menu appearance and behavior could vary.
- Several tools required manual installation and extraction from the Intranet Hacking Tools page.
- Firewall disable steps on all three Windows systems were required for successful scans and were performed as instructed in the sandbox environment.

Reflection:

- This lab tied together multiple enumeration techniques across Windows and Linux toolsets, showing how attackers move from broad discovery to highly specific account and service information.
- The strongest takeaway was that enumeration is dangerous because it bridges passive network visibility and targeted exploitation, making proactive hardening and monitoring essential.